

PES University - Dept. of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #43: Feature Flag & Dynamic Config Manager — Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall evaluate user targeting rules (e.g., User ID hash, Beta cohort) and return accurate boolean feature flag states.	High	Pass: Target 10% rollout cohort receives enabled flag. Fail: User outside cohort receives enabled flag.	Core targeting functionality allowing selective rollout.	Provided sample rule.
FR-002	Functional	The system shall allow Release Managers to configure percentage-based rollout rules for specified application environments.	High	Pass: Feature flag evaluation returns true for exact target percentage of user hashes ($\pm 0.5\%$ variance). Fail: Percentage skew exceeds $\pm 0.5\%$.	Enables safe canary deployments and gradual rollouts.	Required for progressive rollout.
FR-003	Functional	The system shall allow Software Engineers to define environment-specific configuration overrides (e.g., Development, Staging, Production).	High	Pass: Staging receives updated config value while Production retains base value. Fail: Config leaks across environment boundaries.	Prevents unreleased features from affecting live production users.	Supports multi-stage CD pipeline.
FR-004	Functional	The system shall push real-time flag update notifications to connected Client SDKs via Server-Sent Events (SSE) or WebSockets.	Medium	Pass: SDK updates flag cache within 500ms of publish event without application restart. Fail: SDK cache remains stale > 5s or requires restart.	Ensures instant feature toggling without client app polling overhead.	Real-time synchronization constraint.
FR-005	Functional	The system shall log all configuration changes and enable Release Managers to rollback flag state to any previous version.	Medium	Pass: Rollback command restores historical configuration within 3s and records audit trail. Fail: Version state corrupted or audit log missing user ID.	Crucial for rapid incident recovery when bad code is toggled on.	Operational resiliency requirement.
NFR-001	Non-Func (Performance & Security)	The feature flag evaluation API must execute in under 15 ms to minimize overhead when integrated into client critical paths.	High	Pass: Benchmarking tests confirm target latency (<15ms p99) and security standards under simulated peak load. Fail: Latency exceeds 15ms or unauthenticated request succeeds.	Prevents feature evaluation from degrading host application latency.	Provided sample NFR.
NFR-002	Non-Func (Availability & Reliability)	The system shall maintain 99.99% service availability and fallback to local SDK cached rules during network partition events.	High	Pass: Client SDK continues evaluating flags using cached state when API host is unreachable. Fail: SDK throws unhandled exception or blocks thread execution.	Ensures high resilience and zero single-point-of-failure impact on host apps.	High availability & fault tolerance.